

Yingjing Huang

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Research Interest

GIScience, GeoAI, Deep Learning, Computer Vision, Urban Visual Intelligence, Social Sensing

Education

Peking University

Ph.D. in Geographic Information Science

• Advisors: Yu Liu, Fan Zhang

Beijing, China

Sep. 2021 - Jun. 2025

Wuhan University

MS. in Geographic Information Science

• Advisor: Teng Fei

Wuhan, China

Sep. 2018 - Jun. 2021

Wuhan University

BSc. in Geographic Science

• Advisors: Yiyun Chen, Teng Fei

Wuhan, China

Sep. 2013 - Jun. 2017

Experience

University of Vienna

Postdoc

• Advisor: Krzysztof Janowicz

Vienna, Austria

Mar. 2026 - current

Peking University

Research Associate

• Advisor: Fan Zhang

Beijing, China

Jul. 2025 - Feb. 2026

MIT Senseable City Lab

Visiting Ph.D. Student

• Advisors: Carlo Ratti, Fábio Duarte, Simone Mora

Cambridge, US

Jan. 2023 - Jan. 2024

Publications

Publications reported in reverse chronological order

(# co-first author, Google citations = 281)

Ongoing and Submitted Manuscripts

- [1] **Yingjing Huang**[#], Yong Li[#], Fan Zhang, & Yu Liu. Measuring sustainable development with urban imagery. (In Preparing)
- [2] **Yingjing Huang**, Yuqing Wang, Ce Hou, Yanhua Yao, Fabio Duarte, & Fan Zhang. 2026. Measuring urban inequality across private and public spaces. *Cities*. (Under Review)
- [3] **Yingjing Huang**, & Fan Zhang. 2026. Advancing urban sustainability with visual intelligence. *Science Bulletin*. (Under Review)
- [4] Jiapeng Li, **Yingjing Huang**, Fan Zhang, Yu Liu. 2026. StreetTree: A Large-Scale Global Benchmark for Fine-Grained Tree Species Classification. arXiv preprint 2602.19123. (Under Review)

Peer-reviewed Journal Articles

- [1] Yong Li[#], **Yingjing Huang**[#], Fan Zhang. 2026. Learning street view representations based on a spatiotemporal contrastive learning framework. *Computers, Environment and Urban Systems*. (IF = 8.3, Rank in Urban Studies = 3/288, Q1)
- [2] Jing Xiao, Teng Fei, Bo Yu, **Yingjing Huang**, & Yunyan Du. 2025. Street view images help to reveal the impact of noisy environments on the survival duration of stroke patients. *International Journal of*

Health Geographics, 24(1), 27. (IF = 3.2, Rank in Public Health, Environmental and Occupational Health = 106/687, Q1)

- [3] Yu Liu, Xuechen Wang, Yidan Wang, Fei Huang, **Yingjing Huang**, Yong Li, Weiyu Zhang, Shuhui Gong, Gengchen Mai, Yao Yao, Yang Yue, Haifeng Li, & Fan Zhang. 2025. Representation learning for geospatial data. *Annals of GIS*, 1–27. (IF = 3.3, Rank in Earth and Planetary Sciences = 25/198, Q1)
- [4] Jin Zhu, **Yingjing Huang**, Ziyue Cao, Yue Zhang, Yuan Ding, & Jinglong Du. 2025. Evaluating urban greenery through the front-facing street view imagery: Insights from a Nanjing Case Study. *ISPRS International Journal of Geo-Information*, 14(8), 287. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)
- [5] **Yingjing Huang**, Fan Zhang, Lun Wu, and Yu Liu. 2025. Measuring urban physical environments using image deep features. *Cities*, 166, 106196. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)
- [6] **Yingjing Huang**, Fan Zhang, Yong Li, Lun Wu, & Yu Liu. 2025. Intelligent computational representation of urban imagery. *Geomatics and Information Science of Wuhan University*. (In Chinese, a leading journal in Geomatics in China, indexed by EI Compindex)
- [7] **Yingjing Huang**, Rohit Sanatani, Chang Liu, Yuhao Kang, Fan Zhang, Yu Liu, Fabio Duarte, and Carlo Ratti. (2025). No “true” greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment*, 269, 112395. (IF = 7.6, Rank in Geography, Planning and Development = 18/841, Q1)
- [8] **Yingjing Huang**, Fan Zhang, Yong Gao, Wei Tu, Fabio Duarte, Carlo Ratti, Diansheng Guo, and Yu Liu. 2023. Comprehensive urban space representation with varying numbers of street-level images. *Computers, Environment and Urban Systems*, 106, 102043. (IF = 8.3, Rank in Urban Studies = 3/288, Q1)
- [9] Meng Bian, Shuyi Guo, Wei Wang, Yuhui Ouyang, **Yingjing Huang**, and Teng Fei. 2021. Nextday forecast of Beijing pollen concentration fused with vegetation remote sensing data—Implementing a nonlinear autoregressive neural network model with external input. *Journal of Geo-Information Science*, 2021,23(09):1705-1713. (in Chinese, a key academic journal in GIS, indexed in the CSCD Core database)
- [10] **Yingjing Huang**, Teng Fei, Mei-Po Kwan, Yuhao Kang, Jun Li, Yizhuo Li, Xiang Li, and Meng Bian. 2020. GIS-based emotional computing: A review of quantitative approaches to measure the emotion layer of human–environment relationships. *ISPRS International Journal of Geo Information*, 9, 551. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)
- [11] **Yingjing Huang**, Jun Li, Guofeng Wu and Teng Fei. 2020. Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus. *Cities*, 102, 102719. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)
- [12] Yizhuo Li, Teng Fei, **Yingjing Huang**, Jun Li, Xiang Li, Fan Zhang, Yuhao Kang and Guofeng Wu. 2020. Emotional habitat: Mapping the global geographic distribution of human emotion with physical environmental factors using a species distribution model. *International Journal of Geographical Information Science*, 1-23. (IF = 5.1, Rank in Geography, Planning and Development = 31/841, Q1)
- [13] Shuangyin Zhang, Jun Li, Siying Wang, **Yingjing Huang**, Yizhuo Li, Yiyun Chen and Teng Fei. 2020. Repaid identification and prediction of cadmium–lead cross-stress of different stress levels in rice canopy based on visible and near-infrared spectroscopy. *Remote Sensing*, 12, 469. (IF = 4.1, Rank in General Earth and Planetary Sciences = 13/198, Q1)
- [14] Yiyun Chen, Tianci Qi, **Yingjing Huang**, Yuan Wan, Ruiying Zhao, Lin Qi, Chao Zhang, and Teng Fei. 2017. Optimization method of calibration dataset for VIS-NIR spectral inversion model of soil organic matter content. *Transactions of the Chinese Society of Agricultural Engineering*, 06, 107-114. (in Chinese, a top-tier journal in Agricultural Engineering in China, indexed by EI Compindex)

Patents

- [1] Fan Zhang, **Yingjing Huang**, Yu Liu. Method, Device, and Storage Medium for Comprehensive Landscape Representation Based on Street Imagery. China Patent ZL202510142438.1, Granted December

16, 2025.

- [2] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Image Processing Method, Device, Product, Equipment, and Medium. China Patent Application CN116977690A, Published October 31, 2023, Patent Pending.
- [3] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Method and Related Device for Identifying Regions in Space. China Patent Application CN116958803A, Published October 27, 2023, Patent Pending.

Teaching

Geo-Data and Geo-Data Management

Lecture

University of Vienna

Winter 2026

Geographic Information Retrieval

Lecture

University of Vienna

Winter 2026

Conceptual Modelling and Programming

Lecture

University of Vienna

Summer 2026

Spatial Analysis and GIS Applications

Teaching assistant

Peking University

Spring 2025

Grants

2025-2029	Key Program of the National Natural Science Foundation of China (No. 42430106) Spatio-temporal optimization with geographical artificial intelligence	<i>Contributor</i>
2024-2026	National Natural Science Foundation of China (No. 42371468) Representing Urban Physical Environment using Street View Imagery	<i>Contributor</i>
2022-2025	National Natural Science Foundation of China-Excellent Young Scientists Fund Program (Overseas) (No. 41901321) Geospatial Data Mining & Visual Intelligence	<i>Contributor</i>
2022-2023	International Research Center of Big Data for Sustainable Development Goals (CBAS) Key Technologies and Applications of Big Earth Data for Sustainable Development Goals	<i>Contributor</i>
2021-2022	Tencent Academic Research Fund (No. RAGR20210101) Identifying urban informal settlement using multi-source big geo-data	<i>Key Contributor</i>
2020-2022	National Natural Science Foundation of China (No. 41901321) Understanding the relationship between urban physical environment and urban mobility patterns quantitatively using street view imagery	<i>Contributor</i>
2017-2021	National Basic Research Program of China (No. 2017YFB0503602) Geospatial Big Data Mining and Spatiotemporal Pattern Discovery	<i>Contributor</i>

Honors & Awards

	Postdoc-NeT-AI Fellowship on Explainable AI	
2025	A highly competitive fellowship to conduct one week of visiting research in Germany, funded by German Academic Exchange Service (DAAD)	<i>Germany</i>
	Visiting Doctoral Research Scholarship	
2022	A highly competitive fellowship to conduct one year of visiting research abroad, funded by China Scholarship Council (CSC)	<i>China</i>
	“Excellent graduate student”	
2020	National award for the excellent graduate student in Wuhan University	<i>China</i>
	Kwang-Hua Scholarship of Wuhan University	
2020	Scholarship to the excellent student	<i>China</i>
	Second-Class Scholarship of Wuhan University	
2020	Awards to outstanding student in academic in Wuhan University	<i>China</i>
	Third-Class Scholarship of Wuhan University	
2015	Awards to outstanding student in academic in Wuhan University	<i>China</i>

Talks

Invited talks

Analysis methods and applications of street view imagery	<i>Online</i>
Nanjing Normal University, China	<i>Aug. 2025</i>
[Link] (in Chinese)	
Intelligent computational representation of urban imagery	<i>Online</i>
GISChat (A non-profit, GIS knowledge-sharing platform organized by Chinese scholars)	<i>Aug. 2025</i>
[Link] (in Chinese)	

Conference talks

Measuring urban physical environments using image deep features	<i>Nanchang, China</i>
The 26th Inter-University Symposium on Asian Megacities	<i>Sep. 2025</i>
Comprehensive urban space representation with varying numbers of street-Level images	<i>Honolulu, US</i>
American Association of Geographers 2024 Annual Meeting	<i>Apr. 2024</i>
Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus	<i>Beijing, China (online)</i>
2021 International Graduate Workshop on GeoInformatics	<i>Dec. 2021</i>

Conference posters

No “true” greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement	<i>Vienna, Austria</i>
EGU General Assembly 2026	<i>May 2026</i>

Guest Editor

- Transactions in Urban Data Science and Technology

Journal Reviewer

- ISPRS Journal of Photogrammetry and Remote Sensing
- Landscape and Urban Planning
- Cities
- Computers, Environment and Urban Systems
- Building and Environment
- Sustainable Cities and Society
- Geography and Sustainability
- Urban Forestry & Urban Greening
- International Journal of Geographical Information Science
- Environment and Planning B: Urban Analytics and City Science
- Travel Behaviour and Society
- Applied Geography
- Scientific Reports
- IEEE Transactions on Human-Machine Systems
- Journal of Urban Technology
- Computational Urban Science
- Geomatics and Information Science of Wuhan University (in Chinese)
- Journal of Transportation Engineering and Information (in Chinese)

Conference Organization

- Co-organizer, 1st International Workshop on Geography According to Foundation Models, AGILE 2026, Tartu, Estonia, June 16, 2026

Skills

Programming Python, Matlab, R, Javascript, LaTeX

Packages Sklearn, Pytorch

Databases MySQL, SQLite, MongoDB

GIS skills ArcGIS, QGIS, Geopandas, Arcpy

OS Windows, Linux, Mac OS

Languages Mandarin Chinese (*Native*), English (*Advanced*)